Cassava 4 — Configs

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The best cassava system is a Noisy-Student EfficientNet-B0 trained with MixUp/CutMix and per-class focal loss, combined with a ConvNeXt-Base minority-class specialist through a per-class weighted average. Values not listed here are in each notebook's `CONFIG` cell.

Best general model (Exp 10) and specialist (Exp 11)

Setting	Exp 10 — general model	Exp 11 — minority specialist
Backbone	EfficientNet-B0, Noisy-Student weights	ConvNeXt-Base
Classes	All 5	CBB, CBSD, CGM, Healthy (CMD excluded)
Input	384×384, CLAHE	Same
Augmentation	RandomResizedCrop (scale 0.9–1.0), horizontal flip p 0.5, vertical flip p 0.3, transpose p 0.3	Same
Batch mixing	Per batch: 33% CutMix, 33% MixUp, 34% unchanged	Same
Loss	Per-class focal loss (gamma per class in notebook)	Per-class focal loss, gamma = [3.2, 2.2, 1.8, 2.0]
Phase 1	Backbone frozen, `learning_rate`	Same
Phase 2A	Top 30 layers unfrozen, `fine_tune_lr`	Same, half the phase-2 epochs
Phase 2B	Whole network, 0.3 × `fine_tune_lr`	Same, other half
LR schedule	ReduceLROnPlateau on val_loss, factor 0.2, min lr 1e-7	Same; patience 3 / 2 / 2 by phase
Folds	5-fold StratifiedKFold	5-fold StratifiedKFold on the specialist subset (different folds)

Ensemble (Exp 11): per-class weighted average of EfficientNet-B0 (Exp 10) and the specialist: combined[c] = (p_EB0[c] × w_EB0[c] + p_spec[c] × w_spec[c]) / (w_EB0[c] + w_spec[c]). The weights are each model's per-class F1 from training — EfficientNet-B0: 0.63, 0.79, 0.80, 0.96, 0.72; specialist: 0.80, 0.89, 0.93, 0.00, 0.85 (CBB, CBSD, CGM, CMD, Healthy). The specialist's CMD weight is 0, so EfficientNet-B0 alone decides CMD. The 4-class specialist output is expanded to 5 classes with CMD = 0 before combining.

Defaults set by earlier experiments

From	Setting
Exp 1	Class weights = N / (5 × class count); early stopping
Exp 3	Two-phase training: warm-up lr 1e-4 for 15 epochs, then full fine-tune lr 1e-5 for 30 epochs
Exp 4	Offline CLAHE (clip 2.0, 8×8); phase 1 extended to 25 epochs; cosine-decay learning rate over 40 phase-2 epochs, patience 15
Exp 5	Input 224 → 384

Cassava in the cross-dataset study

When the pipeline was rerun on the shared grouped folds for Paper 2, cassava used: ConvNeXt-Base, 384×384, CLAHE, three phases (0.13 M → 28.6 M → 87.7 M trainable parameters), mixed precision, group-aware 10% validation split from the training fold, class weights from the training fold, seed 17 only (about 4 GPU-hours per fold). A control run kept the backbone frozen in all three phases. Full details are in the cross-dataset doc.